

Lab07

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```
library(DBI)
library(RSQLite)

# =====
# 1. Conectar ao banco de dados
# =====
# Ajuste o caminho para onde você salvou o arquivo baixado
conn <- dbConnect(RSQLite::SQLite(), "database.sqlite3")

# =====
# 2. Explorar o esquema do banco
# =====
tabelas <- dbListTables(conn)
tabelas
```

```
[1] "course_offerings"      "courses"              "grade_distributions"
[4] "instructors"           "rooms"                 "schedules"
[7] "sections"             "subject_memberships"  "subjects"
[10] "teachings"
```

```
# Colunas de cada tabela (rode e confira contra o diagrama do enunciado)
for (tb in tabelas) {
  cat("\n---", tb, "---\n")
  print(dbListFields(conn, tb))
}
```

```
--- course_offerings ---
[1] "uuid"          "course_uuid" "term_code"   "name"

--- courses ---
[1] "uuid"  "name"  "number"

--- grade_distributions ---
[1] "course_offering_uuid" "section_number"      "n_count"
[4] "c_count"             "f_count"             "cr_count"
[7] "p_count"             "nw_count"            "other_count"
```

```

[10] "bc_count"          "u_count"          "d_count"
[13] "ab_count"          "a_count"          "s_count"
[16] "i_count"           "b_count"          "nr_count"

--- instructors ---
[1] "id"    "name"

--- rooms ---
[1] "uuid"          "facility_code" "room_code"

--- schedules ---
[1] "uuid"          "start_time" "end_time"    "mon"      "tues"
[6] "wed"           "thurs"      "fri"         "sat"      "sun"

--- sections ---
[1] "uuid"          "course_offering_uuid" "section_type"
[4] "number"        "room_uuid"          "schedule_uuid"

--- subject_memberships ---
[1] "subject_code"    "course_offering_uuid"

--- subjects ---
[1] "code"          "name"          "abbreviation"

--- teachings ---
[1] "instructor_id" "section_uuid"

```

```

# =====
# 2. Professores que lecionaram disciplinas de Estatística
# =====
# subjects.abbreviation = 'STAT' identifica o assunto
# subject_memberships liga subjects a course_offerings
# sections liga course_offerings às seções (turmas)
# teachings liga seções a instrutores
# instructors traz o nome do professor

query_professores_stat <- "
SELECT DISTINCT i.id, i.name
FROM instructors AS i
INNER JOIN teachings AS t      ON t.instructor_id = i.id
INNER JOIN sections AS s      ON s.uuid = t.section_uuid
INNER JOIN course_offerings AS co ON co.uuid = s.course_offering_uuid
INNER JOIN subject_memberships AS sm ON sm.course_offering_uuid = co.uuid
INNER JOIN subjects AS sub    ON sub.code = sm.subject_code
WHERE sub.abbreviation = 'STAT';
"

```

```
professores_stat <- dbGetQuery(conn, query_professores_stat)
professores_stat      # quem são
```

	id	name
1	4392906	MINJING TAO
2	393380	DONALD PORTER
3	4146030	SHENG WANG
4	5697552	KUNLING HUANG
5	4898014	DONG XIA
6	4613728	JENNIFER NGUYEN
7	5520811	SEULKEE YUN
8	5516726	DUZHE WANG
9	5696436	YONGFENG WU
10	4636315	JARED BROWN
11	5700058	YUCHEN ZHOU
12	4641200	VICTOR LUO
13	4884161	GUN WOONG PARK
14	5011312	BROOK LUERS
15	4728844	DUY NGUYEN
16	608997	JOHN GILLET
17	4161984	QUOC TRAN
18	4576059	YAOYAO XU
19	4436368	JIE ZHANG
20	5015113	LONG PHAN
21	5161294	WESLEY CHANG
22	5329574	TONG LI
23	982957	SHANE HUBLER
24	4437355	FAN YANG
25	4731731	JU HEE CHO
26	4890283	JEEA CHOI
27	4885477	SHIXUE LIU
28	2602108	WEI-YIN LOH
29	4155990	BIN DAI
30	4145889	XIPEI YANG
31	2602050	DOUGLAS M. BATES
32	783769	MICHAEL GEORGE ILTIS
33	4845798	LIAM JOHNSTON
34	5498648	CRYSTAL CHEN
35	5693277	YIFAN MEI
36	5690182	RUI CHEN
37	4734274	JILI WANG
38	5017774	LUWAN ZHANG
39	5017655	GINA BENNINGER
40	5410648	JIE SONG
41	3791802	XIWEN MA
42	3456935	HAODA FU
43	3773167	DEYUAN JIANG

44	3973318	JINGJIANG PENG
45	3566378	ERICA LEE DEADMAN
46	3775809	AKICHIKA OZEKI
47	4280220	MENG SONG
48	5520701	HAO TENG
49	4897725	DONGGYU KIM
50	5332940	COLIN LONGHURST
51	361804	FREDERICK BOEHM
52	5518378	YONGSU LEE
53	5320625	GONZALO CONTADOR
54	5514514	HAO CHEN
55	4588103	XIN ZHANG
56	5632570	BO YANG
57	5898424	RUNGANG HAN
58	5068296	SEAN KENT
59	6043299	DEBRAJ DAS
60	5327139	YI LI
61	5887049	CLAIRE BOBST
62	5883206	CHAN PARK
63	5323628	SIJING LI
64	5017571	SHENGJI JIA
65	5903922	YUCHANG WU
66	3626007	ZHUANG WU
67	5170801	XIAOMAO LI
68	5321537	HYEBIN SONG
69	5326059	YOURAN QI
70	5177493	YOUNG MIN PARK
71	5186047	BIN ZHANG
72	4582107	QIURONG CUI
73	4733192	SEHO PARK
74	4285558	SHENG ZHANG
75	5168069	HUIKUN ZHANG
76	5162339	ALYSSA DIGILIO
77	5214858	YUQING XU
78	4538540	STEPHEN BERG
79	5326263	PEIGEN ZHOU
80	4721620	YAN CHEN
81	4460098	QIONG ZHANG
82	4278664	YOUNGDEOK HWANG
83	5010171	HAN CHEN
84	640407	NICHOLAS STEPHEN KEULER
85	3107260	ZHENGXIAO WU
86	4144479	YUJIN CHUNG
87	4590481	WENWEN ZHANG
88	4149787	PERLA REYES
89	3974100	QI TANG
90	3629214	CHIA-CHIEH LIN
91	4433317	JIALE XU

92	4285354	YANG ZHAO
93	3620612	XINWEI DENG
94	4133622	MING XIE
95	3783649	JINGCI MENG
96	143932	ISMOR FISCHER
97	4590715	JIAJIE CHEN
98	4138542	ZUOFENG SHANG
99	4435790	HAO ZHENG
100	4590559	XU XU
101	4436066	BIN ZHU
102	4432994	QUEFENG LI
103	3961448	YU QIU LIU
104	3981051	JUN LI
105	5204089	FAN GAO
106	4788841	VICTORIA MANSFIELD
107	5498092	ANQI SHI
108	5317373	KYLE HEBERT
109	5322566	XIUYU MA
110	4597046	CHENLIANG XU
111	3976099	CHANHAN HSU
112	4906458	YUANZHI LI
113	2602032	KAM-WAH TSUI
114	4430246	XINYU SONG
115	4879129	TIMOTHY IDOWU
116	4905505	XIAO NIE
117	4586558	CHEN CHENG
118	5003565	ANDREW LESLIE
119	4895244	YAOGUO XIE
120	4744350	KEVIN PACKARD
121	4586128	YI LIU
122	5322094	WENZHI CAO
123	4610245	JOSEPH DEUTSCH
124	5521838	LAN LUO
125	3727371	RYAN ZEA
126	2601091	ROBERT WARDROP
127	3981781	SANGBUM CHOI
128	4285528	XIN LI
129	3573852	SEUNGBONG HAN
130	3339239	ZHIGUO XIAO
131	2601613	RICHARD A. JOHNSON
132	4128750	RUI TANG
133	801445	IAN BRANSTAD RILEY
134	3484236	BO HUANG
135	4149843	HEATHER MARIE BRAZEAU
136	4286138	XU HE
137	495087	GARY HOWARD SCHROEDER
138	3781764	JUN ZHANG
139	3973813	CHENXI LI

140	3955531	SONA Z SWANSON
141	3791787	YUAN JIANG
142	3620552	JUNHEE HAN
143	4578330	LILUN DU
144	5170924	XIRAN WANG
145	5099221	ZIFENG ZHAO
146	5176022	CUIZE HAN
147	5170492	YING ZHANG
148	4880968	JOHN DAVIS
149	5018463	CHEN JING
150	4902482	JINGLAN LI
151	4285538	YI CHAI
152	4904255	THU LE
153	4958796	ALAN HUANG
154	4725072	HAOYANG FAN
155	5690247	BOWEN HU
156	5664494	TZU HSIANG HUNG
157	5567363	TUN LEE NG
158	4787911	ABIGAIL BENZINE
159	5656034	ALEXANDER COVINGTON
160	3376532	CHELSEY GREEN
161	5708167	MICHAEL KUTZLER
162	5693114	LILI ZHENG
163	5488796	REBECCA POST
164	5010560	LU YANG
165	5020061	RUOSI GUO
166	3668575	XUN ZHAO
167	3965653	MIN NIU
168	4603165	JIAN WU
169	3780817	WEI ZHENG
170	4726666	XINJIE HE
171	4292460	JIE WEI
172	4721578	CLAUDIA SOLIS LEMUS
173	4726703	XUEYAO CHEN
174	5302154	MANJUSHA KANCHARLA
175	4947463	MITCHELL PAUKNER
176	5258581	XIAOWU DAI
177	5906594	SHENG WANG
178	5592111	CHENGNING ZHANG
179	5862942	GINA OH
180	5192960	LUXI CAO
181	4953246	SONG WANG
182	4435446	LAM HO
183	4896939	TIEN VO
184	4598772	SHIZHEN WANG
185	5176905	SOHEIL SADEGHI
186	5144884	BINGYING XIE
187	4957688	GENG LI

188	4636145	JENNIFER BIRSTLER
189	4195000	MARIA KAMENETSKY
190	5478075	KYLE HERBET
191	4886987	JIANCHANG HU
192	4896172	TRAM TA
193	4733966	YUAN WANG
194	3777992	KAZUHIKO SHINKI
195	3622809	SIJIAN WANG
196	3613299	CHIEN-WEI CHEN
197	5852265	BEHZAD AALIPUR
198	4264265	DEREK BEAN
199	4140469	BEN ADAM HAALAND
200	2824494	ELOISA D CHAVAS
201	3800708	YALI WANG
202	3959251	VARSHA KULKARNI
203	3253489	SCOTT JOSEPH HETZEL
204	3575433	MIN JUNG LEE
205	3460189	FANG FANG
206	4199106	KRISTEN CYFFKA
207	3351080	TING-LI LIN
208	751075	BRET LARGET
209	4491860	KATHERINE GOODE
210	2602042	ERIK NORDHEIM
211	5523214	NING FAN
212	4646729	ANRU ZHANG
213	5214164	GARVESH RASKUTTI
214	5476019	KRISHNAKUMAR BALASUBRAMANIAN
215	4695492	YAZHEN WANG
216	4712098	CHAOYANG YU
217	5323691	YILIN ZHANG
218	4108525	ZHENGJUN ZHANG
219	4194670	XIUFENG SHAO
220	5167973	YUAN LI
221	4753018	BRET HANLON
222	3980675	LILI LAN
223	3955546	CECILE ANE
224	1426304	ZHANG CHUNMING
225	3443667	LEI XU
226	5514561	MUHONG GAO
227	4963378	AUGUST JENSEN
228	4591139	XIAO GUO
229	2601039	THOMAS G. KURTZ
230	3523865	NICHOLAS HENDERSON
231	4903489	CHENSHENG KUANG
232	4590738	TAERI UHM
233	3882760	NORBERT BINKIEWICZ
234	4929500	KARL ROHE
235	3095399	JUNGWON MUN

236	1126199	JOSEP GINEBRA
237	3781422	TAO YU
238	4286679	XINXIN YU
239	5323690	HAO ZHOU
240	4277002	LIE XIONG
241	3338237	QIAN ZHIGUANG
242	5515201	SHANG WU
243	5905014	SHAN LU
244	3122658	STEPHEN AARON STANHOPE
245	3103459	LANCINE KONATE
246	4886028	GUILHERME VIEIRA NUNES LUDWIG
247	5631398	PAUL SAVARIAPPAN
248	5015194	CUICUI QI
249	3404299	KJELL DOKSUM
250	5305675	DEREK NORTON
251	4261114	KEEGAN KORTHAUER
252	5012817	KAILEI CHEN
253	5023414	THEVAASIINEN CHANDERENG
254	3806328	JOHN KANE
255	4593965	SOKOL VAKO
256	5718599	QING LI
257	5514517	YOUNG LEE
258	4734607	HYUNSEUNG KANG
259	5694563	BEHZAD AALIPUR HAFSHEJANI
260	5170885	SHULEI WANG
261	5693629	ZHONGJIE YU
262	5514932	LIN QI
263	4286103	JIWEI ZHAO
264	3801422	SUNDUZ KELES
265	4588354	SHUYUN YE
266	3483534	HUI WANG
267	3978606	KEVIN HASEGAWA ENG
268	3343005	XIAODAN WEI
269	5167906	MARIA ROJO
270	3343055	RUIFENG XU
271	5644228	YONGJOON KIM
272	232828	NELLIE LAUGHLIN
273	270258	MICHAEL HOGAN
274	4957916	TIMOTHY IDOWU
275	4541610	KARL BROMAN
276	789363	REBECCA KOSCIK
277	5887858	RONGJUN ZHU
278	5906523	BI CHENG WU
279	5907599	YUNONG LIN
280	4588525	ZHIGENG GENG
281	4729481	XIAOPING FENG
282	4271924	SHUANG HUANG
283	4436002	GUANNAN SUN

284	4140397	JEE YEON KIM
285	5127444	GREGORY SHINAULT
286	3230527	TONGHAI YANG
287	3374880	TIMO SEPPALAINEN
288	4538519	BENEDEK VALKO
289	2602109	DAVID GRIFFEATH
290	4308515	DAVID ANDERSON
291	1515839	PHILIP WOOD
292	4601948	FLORIAN BERTRAND
293	3956063	JASON RICHARD SWANSON
294	3573775	ALEXANDER KISELEV
295	4752238	SAMUEL STECHMANN
296	4434130	WAI TONG FAN
297	3620114	JUN YIN
298	2600973	STEFFEN LEMPP
299	3987897	RUIFANG SONG
300	2601633	JAMES D KUELBS
301	129415	ANATOLE BECK
302	5620719	DONGHYUN LEE
303	3674168	ANDREJ ZLATOS
304	5846795	DANIELE CAPPELLETTI
305	6029366	MIHAELA IFRIM
306	5110390	SEBASTIEN ROCH
307	5354849	SCOTT HOTTOVY
308	1634418	JONATHON PETERSON
309	2601051	STEPHEN WAINGER
310	4680082	ALEXANDER FISH
311	4761200	GREGORIO MORENO-FLORES
312	3117835	SUKHENDU MEHROTRA
313	2600233	DIETRICH UHLENBROCK
314	3437432	LEV BORISOV
315	2600141	RICHARD A BRUALDI
316	2600893	ARNOLD MILLER
317	3467530	MATTHEW BALLARD
318	3661782	ALBRECHT KLEMM
319	2601093	PAUL M TERWILLIGER
320	5452538	STEVEN SAM
321	3127329	KEN ONO
322	4957895	ROBERT HARRON
323	4172260	YANNAN QIU
324	4896511	JOHN WILTSHIRE-GORDON
325	207204	KYUNGMAH KIM
326	3855025	CHRISTOPHER WAGNER
327	967335	PAUL RATHOUZ
328	3382451	STEPHEN WRIGHT
329	4632211	BENJAMIN RECHT
330	1114818	MICHAEL FERRIS
331	3627934	JESSE THOMAS HOLZER

332	5167990	HUILIN HU
333	5451706	ALBERTO DEL PIA
334	2601512	ROBERT R MEYER
335	3127330	SHI JIN
336	333334	MARY LINDSTROM
337	3111842	ADIN-CRISTIAN ANDREI
338	5840930	MICHAEL LIOU
339	4724058	CHAOQUN MEI
340	340544	CHRISTINA M. KENDZIORSKI
341	893827	RONALD GANGNON
342	3780010	ROBERT WAYNE GREEN
343	4312759	COLE COOK
344	2601621	MURRAY CLAYTON
345	4633355	QI JIANG
346	321374	JUN ZHU
347	4952769	ANNE BRUCKNER
348	5520995	TING YE
349	1079018	JUN SHAO
350	3623432	HUAIBAO FENG
351	5591917	ZIJIAN NI
352	417071	THOMAS COOK
353	2601926	RICHARD J. CHAPPELL
354	3404298	MOO K CHUNG
355	2601916	MICHAEL NEWTON
356	2602055	BRIAN YANDELL
357	394810	MICHELLE HARRIS
358	2600689	NORMAN DRAPER
359	3010828	YU MENGANG
360	3077702	MING YUAN
361	3783301	FANGFANG WANG
362	4541633	JUNHO LEE
363	2600488	GRACE WAHBA
364	3011936	JASON P FINE
365	989414	MICHAEL RENE KOSOROK
366	4426932	YAJUAN SI
367	4726678	LU MAO
368	2601596	DAVID DEMETS
369	5515948	JAMES ANDERSON
370	5189535	EDWARD ERKER
371	4699427	SHUAI CHEN
372	2601674	MARI PALTA
373	4272783	GUANHUA CHEN
374	3884290	SRIKANTHMADHAVAN ARAVAMUTHAN
375	1505727	NATALIA DE LEON GATTI
376	1505756	GUILHERME ROSA
377	3457484	YANBING ZHENG
378	138086	CHRISTINE SORKNESS
379	245914	MARIAN R FISHER

```
nrow(professores_stat)      # quantos são
```

```
[1] 379
```

```
# =====  
# 3. Nota média ponderada (GPA) por oferecimento  
# =====  
# Escala: A=4, AB=3.5, B=3, BC=2.5, C=2, D=1, F=0  
# grade_distributions guarda as contagens de alunos em cada conceito,  
# por course_offering_uuid + section_number (+ section_type)  
  
query_gpa <- "  
SELECT  
    i.id            AS instructor_id,  
    i.name          AS instructor_name,  
    c.number        AS course_number,  
    c.name          AS course_name,  
    co.uuid         AS course_offering_uuid,  
    gd.a_count, gd.ab_count, gd.b_count, gd.bc_count,  
    gd.c_count, gd.d_count, gd.f_count  
FROM grade_distributions AS gd  
INNER JOIN course_offerings AS co    ON co.uuid = gd.course_offering_uuid  
INNER JOIN courses AS c            ON c.uuid = co.course_uuid  
INNER JOIN subject_memberships AS sm ON sm.course_offering_uuid = co.uuid  
INNER JOIN subjects AS sub         ON sub.code = sm.subject_code  
INNER JOIN sections AS s           ON s.course_offering_uuid = co.uuid  
                                   AND s.number = gd.section_number  
INNER JOIN teachings AS t          ON t.section_uuid = s.uuid  
INNER JOIN instructors AS i        ON i.id = t.instructor_id  
WHERE sub.abbreviation = 'STAT';  
"  
  
notas <- dbGetQuery(conn, query_gpa)  
  
# Força conversão para numérico (evita erro de tipo character)  
cols_contagem <- c("a_count", "ab_count", "b_count", "bc_count",  
                  "c_count", "d_count", "f_count")  
notas[cols_contagem] <- lapply(notas[cols_contagem], as.numeric)  
  
head(notas)
```

	instructor_id	instructor_name	course_number
1	393380	DONALD PORTER	224
2	393380	DONALD PORTER	224
3	4898014	DONG XIA	224

4	4636315	JARED BROWN	224
5	4898014	DONG XIA	224
6	608997	JOHN GILLET	224

		course_name	course_offering_uuid
1	Introductory Statistics for Engineers	0a00ed96-3293-322f-ba23-039f67ce2fa7	
2	Introductory Statistics for Engineers	0a00ed96-3293-322f-ba23-039f67ce2fa7	
3	Introductory Statistics for Engineers	8f3800fe-8d3f-392d-84a9-9509e888f6ac	
4	Introductory Statistics for Engineers	8f3800fe-8d3f-392d-84a9-9509e888f6ac	
5	Introductory Statistics for Engineers	8f3800fe-8d3f-392d-84a9-9509e888f6ac	
6	Introductory Statistics for Engineers	03a155a1-32c6-3411-b388-9e59cea81c2c	

	a_count	ab_count	b_count	bc_count	c_count	d_count	f_count
1	17	14	27	7	20	2	0
2	21	16	23	6	13	1	2
3	15	11	27	7	11	3	0
4	11	15	23	20	8	7	0
5	15	3	28	6	11	1	2
6	27	5	13	14	17	3	2

```
# Quantidade de alunos e soma de pontos (numerador da média ponderada)
notas$n_alunos <- with(notas, a_count + ab_count + b_count + bc_count +
                        c_count + d_count + f_count)

notas$soma_pontos <- with(notas, a_count * 4 + ab_count * 3.5 +
                          b_count * 3 + bc_count * 2.5 +
                          c_count * 2 + d_count * 1 +
                          f_count * 0)

notas <- subset(notas, n_alunos > 0) # remove linhas sem alunos avaliados
notas$gpa_oferecimento <- notas$soma_pontos / notas$n_alunos

# =====
# a/b. Professor mais difícil / mais fácil
# =====
# Agrega por professor, ponderando pelo número de alunos em cada turma
por_professor <- aggregate(cbind(soma_pontos, n_alunos) ~ instructor_id +
                           instructor_name,
                           data = notas, FUN = sum)
por_professor$gpa_medio <- por_professor$soma_pontos / por_professor$n_alunos

# GPA menor = disciplina/professor mais difícil; GPA maior = mais fácil
professor_mais_dificil <- por_professor[which.min(por_professor$gpa_medio), ]
professor_mais_facil <- por_professor[which.max(por_professor$gpa_medio), ]

professor_mais_dificil
```

	instructor_id	instructor_name	soma_pontos	n_alunos	gpa_medio
70	2601633	JAMES D KUELS	105.5	40	2.6375

professor_mais_facil

	instructor_id	instructor_name	soma_pontos	n_alunos	gpa_medio
63	4272783	GUANHUA CHEN	64	16	4

```
# =====
# c/d. Disciplina mais difícil / mais fácil
# =====
por_disciplina <- aggregate(cbind(soma_pontos, n_alunos) ~ course_number +
                             course_name,
                             data = notas, FUN = sum)
por_disciplina$gpa_medio <- por_disciplina$soma_pontos /
por_disciplina$n_alunos

disciplina_mais_dificil <-
por_disciplina[which.min(por_disciplina$gpa_medio), ]
disciplina_mais_facil <-
por_disciplina[which.max(por_disciplina$gpa_medio), ]

disciplina_mais_dificil
```

	course_number	course_name	soma_pontos
n_alunos			
26	431	Introduction to the Theory of Probability	6439.5
2231			
gpa_medio			
26			2.886374

disciplina_mais_facil

	course_number	course_name	soma_pontos	n_alunos	gpa_medio
7	628	Data Science Practicum	48	12	4

```
# =====
# 4. Desconectar do banco de dados
# =====
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-22 20:40:19 -03"
```